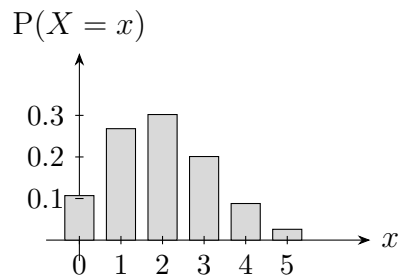


4.19 Quiz: Normal and Binomial Distribution

1. A market stall sells bags of apples. It is known that 20% of the apples have bruises. A bag contains 10 randomly selected apples. Let X be the number of bruised apples in the bag.



- (a) Find the expected number of bruised apples in a bag, $E(X)$. [1]
- (b) Find the probability that none of the apples are bruised. [2]
- (c) Find $P(X = 3)$. [2]
- (d) Find $P(X \leq 3)$. [2]

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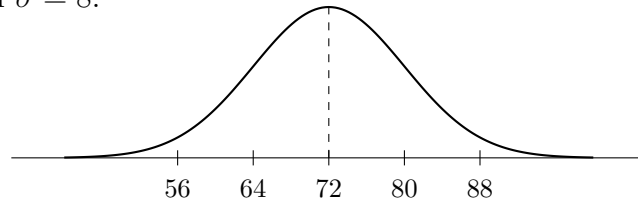
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2. The marks on a mathematics test follow a normal distribution with mean $\mu = 72$ and standard deviation $\sigma = 8$.



- (a) Find the probability that a randomly selected student scores higher than 85. [2]
- (b) Find the probability that a student's mark is between 60 and 80. [2]
- (c) 75% of the students score less than h marks. Find h . [3]

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Answer this section on a separate piece of lined paper

3. The waiting time T (minutes) at a bus stop follows a normal distribution with mean 15 minutes and standard deviation σ minutes. It is known that 4% of waits are longer than 20 minutes.
- (a) Find the value of σ . [3]
- (b) Find $P(T < 12)$. [2]
- (c) On a particular day, 10 people wait at this bus stop. Find the probability that at most 2 of them wait more than 20 minutes. [3]
- (d) A person is selected at random. Given that they waited less than 15 minutes, find the probability that they waited less than 10 minutes. [3]